

# Bryan Nguyen

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<https://bryannguyen.github.io/bryannguyen.github.io/>

714-710-2342

## Education

### University of California, Irvine

Mechanical Engineering, B.S. - GPA: 3.5

Expected Grad: Jun 2027

### Mount San Antonio College

Associate in Science for transfer - Mechanical Engineering & Mathematics

Sept 2022 - Jun 2025

## Skills

**CAD:** SolidWorks, AutoCAD, Fusion 360, OnShape, KiCad

**Manufacturing:** 3D Printing, Laser Cutting, Milling & Lathe Operation, Wood Working, GD&T, Prototyping

**Electronics:** Arduino, Raspberry Pi, Electronic Circuits, Wiring Schematics, Soldering

**Programming:** C++, Python, MATLAB, Simulink

## Projects

### Robotic Manipulation System (LeRobot)

Jan 2026 - Present

- Configured teleoperation using the LeRobot framework to control a follower arm and collect demonstration data
- Trained a task-specific control policy from teleoperated demonstrations to perform object grasping and placement
- Deployed and evaluated the learned policy on hardware, successfully executing block pick and place into a target bin

### Ball Balancing Robot

Dec 2025 - Jan 2026

- Built a 3-DOF delta robot which balances a ball on a platform, using SOLIDWORKS and 3D-printed components
- Implemented real-time ball tracking in Python via OpenCV, processing camera frames to compute ball position error
- Developed inverse kinematics and PID control algorithms in Python to convert vision based position error into joint angle commands for servo motors via serial communication with Arduino board

### 5-DOF Robotic Arm

Jul 2025 - Sept 2025

- Designed, prototyped, fabricated and tested a 5-DOF robotic arm, delivering a payload capacity of 250g
- Created full mechanical assembly in SOLIDWORKS and 3D printed structural parts and custom gears using ABS & PLA
- Built custom electronics and power system utilizing voltage regulation, NEMA-17 base rotation, and servos for joints
- Implemented C++ firmware on an Arduino mapping potentiometer inputs to servo joints for real-time teleoperation

### Propulsion Lead Engineer

Jan 2025 - Jun 2025

- Led a 5-person engineering team in the design, fabrication, and static testing of a solid rocket motor, achieving 2000 N peak thrust and 5776 Ns total impulse
- Designed propulsion hardware in OnShape with emphasis on manufacturability, structural integrity, and assembly
- Performed static tests collecting thrust and burn rate data, analyzed results and modeled performance with MATLAB
- Integrated the propulsion system into a dual deployment rocket and validated performance through flight testing, reaching altitudes over 7000 ft

## Experience

### Undergrad Researcher - Cal Poly ENGAGE Program

Sept 2024 – May 2025

- Designed and executed 6 instrumented static fire experiments, collecting thrust data using a 500 lbf load cell and Arduino based DAQ system sampling at ~ 80 Hz
- Processed and analyzed 3,000+ data points per test in MATLAB to extract thrust, burn time, and impulse
- Performed comparative analysis across 4 system configurations, quantifying performance changes and identifying 35–38% impulse reductions caused by material and process variations

### Rocket Workshop - Lead

Sept 2024 - Jan 2025

- Designed and co-led a 10-week workshop for 30 students, covering CAD, simulation, and fabrication of rockets
- Organized meetings, budgets, and logistics, adapting to challenges like relocating fabrication space
- Provided hands-on mentorship, guiding students through design challenges and technical skills